

Revised Generation Interconnection System Impact Study Report

For Queue Project Q17-0800

Alvin 138 kV
July 25, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q17-0800. The Interconnection Customer proposes a 74.3 MW solar photovoltaic facility interconnecting at the Alvin 138 kV substation in the Virginia area. The request was received on May 19, 2017. The study evaluated system intact and single contingency conditions using the regional planning case, including 230 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 20 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$48,672,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Revital – Crozet 138 kV	New London – Declo 138 kV	165.0	167	5.3
Revital – Dale Mary 138 kV L	Declo – New London 138 kV	145.5	193	5.3
Tes Filer City Station Substation – Bridge River 138 kV	Urban – American 345 kV L	144.5	223	7.8
Cedarville Jct – Detour 138 kV L	St Louis – Nichols 138 kV Line	144.2	248	21.2
Page – Cedarville Junction 138 kV L	St Louis – Nichols 138 kV	143.7	252	21.2
Purdy – Page Substation 138 kV L	St Louis – Nichols 138 kV	143.5	261	21.2
Dale Mabry – Grottoes 138 kV	Declo – New London 138 kV L	138.3	215	5.3
Alvin – Nuclear Fuels 138 kV line	Oak Crk Station – Sans Souci 138 kV	134.0	201	23.0
Mount Jackson – Cherry 138 kV	Chatom – Purdy 138 kV line	124.0	178	7.3
Siloam Springs – Detour 138 kV	Fort Benton – Detour 138 kV L	121.7	181	13.5
Cherry – Dpc Prairie Du Chien Substation 138 kV	Purdy – Chatom 138 kV L	119.7	192	7.3
Mcauley – Crozet 138 kV	Declo Substation – New London 138 kV	119.5	211	5.3
Roosevelt Rd – Elkton D.P 138 kV	Nuclear Fuels Substation – St Louis 138 kV line	110.5	197	14.5
Ft Beton – Detour 138 kV	Siloam Springs – Detour 138 kV	109.4	250	14.6
Houston – Elkton D.P 138 kV	Nuclear Fuels – St Louis 138 kV	109.4	199	14.5

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Wadsworth – Roosevelt Rd Substation 138 kV	St Louis – Nuclear Fuels 138 kV L	108.7	208	14.5
Siloam Springs – Lewisburg 138 kV	Detour – Fort Benton 138 kV	108.6	246	5.8
Nuclear Fuels – St Louis 138 kV	Page – Purdy 138 kV	105.4	192	21.7
Declo – New London 138 kV	St Louis – Nichols 138 kV	101.6	364	5.4
Mount Jackson – Puddledock Rd 138 kV	Purdy – Chatom 138 kV	101.4	206	7.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Revital Substation – Crozet 138 kV	Rebuild line with higher-capacity conductor	\$5,645,000	8% (\$422,000)
Revital – Dale Mary 138 kV L	Rebuild line with higher-capacity conductor	\$6,071,000	8% (\$454,000)
Tes Filer City Station Substation – Bridge Riv 138 kV	Rebuild line with higher-capacity conductor	\$4,756,000	36% (\$1,690,000)
Cedarville Jct – Detour 138 kV L	Rebuild line with higher-capacity conductor	\$14,629,000	39% (\$5,687,000)
Cedarville Jct – Page 138 kV	Rebuild line with higher-capacity conductor	\$11,046,000	39% (\$4,294,000)
Page – Purdy 138 kV	Rebuild line with higher-capacity conductor	\$2,466,000	39% (\$959,000)
Dale Mabry Substation – Grottoes 138 kV	Rebuild line with higher-capacity conductor	\$4,537,000	8% (\$339,000)
Nuclear Fuels – Alvin 138 kV line	Rebuild line with higher-capacity conductor	\$6,925,000	29% (\$1,996,000)
Cherry – Mount Jackson 138 kV	Rebuild line with higher-capacity conductor	\$7,435,000	100% (\$7,435,000)
Detour – Siloam Springs 138 kV	Rebuild line with higher-capacity conductor	\$12,902,000	100% (\$12,902,000)
Cherry – Dpc Prairie Du Chien Substation 138 kV	Rebuild line with higher-capacity conductor	\$6,549,000	100% (\$6,549,000)
Mcauley – Crozet 138 kV	Rebuild line with higher-capacity conductor	\$12,458,000	8% (\$931,000)
Roosevelt Rd – Elkton D.P 138 kV	Rebuild line with higher-capacity conductor	\$894,000	18% (\$158,000)
Ft Beton – Detour 138 kV	Reconductor line	\$1,657,000	62% (\$1,022,000)
Houston – Elkton D.P 138 kV	Reconductor line	\$681,000	18% (\$121,000)
Wadsworth – Roosevelt Rd 138 kV	Reconductor line	\$2,584,000	18% (\$458,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Siloam Springs – Lewisburg 138 kV	Reconductor line	\$1,848,000	39% (\$728,000)
St Louis – Nuclear Fuels 138 kV	Reconductor line	\$1,717,000	41% (\$710,000)
Declo – New London 138 kV	Reconductor line	\$15,384,000	2% (\$386,000)
Mount Jackson – Puddledock Rd 138 kV	Reconductor line	\$1,431,000	100% (\$1,431,000)
	Total allocated to project		\$48,672,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.